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Photo: Erwin Binner

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The discovery of *Melanotaenia jakora* and hints for keeping and breeding

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On 11 August 2017, we drove from Bintuni to Meyado early in the morning in a four-wheel-drive pickup truck. Our tour group, consisting of Henni Ohee, Marten Luter Salossa, Wim Heemskerk, Gary Lange, our local companion Raymond and I, were on our first trip to the southern area of the Vogelkop Peninsula of New Guinea. Coming from Sorong, we had flown to Bintuni in a small plane (Cessna Caravan) and surprisingly quickly found a car and driver there. We had Marten and Raymond to thank for this, as they had kept their mobile phones glowing before departure and then immediately after arrival.

Shortly after Bintuni, the asphalt road became a gravel road. The condition of the road is affected by rainfall and, as it had not rained in the previous few days, we made good progress. We reached the small town of Meyado after only a few hours at around midday. Meyado is a typical Indonesian settler town and there were almost no indigenous Papuans to be seen. In conversation we learned that most of the people living here originally came from Sulawesi, moving over through the Indonesian transmigration program.

We met the mayor and he took us to a house where we could spend the night. This house belonged to an elderly Indonesian couple who lived in a house at the rear and made the newly built front house available to us.



Taking a rest after the long drive.

Photo: Henni Ohee

By the early afternoon we reached our destination for the day but there were still several hours of daylight left, so we asked the mayor for suggestions of small roads that were not marked on the maps. He knew one and came with us and we drove south from Meyado on a small, narrow road that had obviously been built only recently.



Getting ready for fishing.

Photo: Henni Ohee



Raymond and Johannes catching the new species.

Photo: Gary Lange

Several small rivers crossed the road and we stopped to look for fish. At the first spot we couldn't use a seine because of the deep water, the steep bank and a lot of branches in the water, so we fished from the shore with the frame net with great success. We caught lots of small rainbowfish of about 1 cm in length and bagged some of them up. Later they turned out to be a typical Birdsheed species and we named them *Melanotaenia* spec. "Meyado Village".

We continued south and came to a place where a stream ran under the road through a pipe. On one side of the road the stream bed had been enlarged like a pond by dredging work. The water was deep black with humic substances with a visibility of only about 5 cm. The water hardness was too low to measure, the conductivity was only 40 μ S and the water temperature was 26°C. Our local companions called the stream "Kali Jakora".

The dredging work must have been done just recently as the entire area consisted of slippery clay and was not yet overgrown with plants. But I really wanted to get in there because I suspected that there might be other fish species than in "normal" biotopes because of the black water. Only Raymond accompanied me, everyone else stood up on the road and watched us trudge through the clay trying to catch fish with the frame nets. Well, the fishes population density wasn't very high there, but after a few tries we finally had success: small rainbowfishes, clearly something different from what we had caught so far! Almost black in body colour, but iridescent blue in the sunlight!

Raymond and I struggled to catch enough fish. It's quite an effort when you keep getting stuck in the clay and have to pull your feet out again and again. It took a while, but after about an hour we had caught enough specimens. In the bag we could already see that this rainbowfish was a new species. We named it *Melanotaenia* spec. "Kali Jakora".

We then drove to the end of the road to Jakora village, situated on the bank of the Sungai Sebjar (Sebjar River) at the confluence of



Wim and Raymond fishing in a creek near Meyado. Photo: Marten L. Salossa



Female *Melanotaenia jakora*.

Photo: Gary Lange

Kali Jakora. Sungai Sebjar looked very much like it would contain crocodiles, and the people from the village confirmed that their dogs would frequently go missing. So, we drove back very satisfied with what we had caught.

The fish survived the rest of our Birdshead trip in breather bags without any problems. While we had the original black water in the bags at the beginning, we replaced it over time by changing the water with soft water until the fish were fully purged.



Raymond and Johannes in action.

Photo: Gary Lange

The subsequent transport to Jakarta and on to Germany were also successful, and the fishes were finally able to settle in the aquarium. After only a few weeks, it became clear that this species represents a very distinct lineage that has nothing to do with the other rainbowfish species of the Vogelkop Peninsula (Vogelkop clade, as defined by Kadarusman in his 2012 article about rainbowfish clades). This was confirmed a little later by DNA analyses by Dr Peter Unmack of the University of Canberra. This species is not directly related to any other rainbowfish species. A resemblance in the broadest sense exists with *M. gracilis*, but even with this there are considerable morphological differences. Particularly striking is the colouration, which is unique with the pink-white fin margins and the blue glow over the body.

The first spawning occurred after only a few weeks in a spawning mop made of synthetic yarn, however, the eggs turned white and died. I kept the species in my usual aquarium water (slightly hardened rainwater), in which the adults felt very comfortable and showed full colours.

I remembered the water conditions at the place of discovery, and I changed to pure rainwater and added a huge amount of humic substances (Leonardite) to the water so that it became deep black and the fish were no longer visible. I left them in it for 10 days with a spawning mop, then caught the adults out and put them in another tank. On the second day the first larvae could be seen which I was able to rear without much difficulty on live infusoria and later with brine shrimp nauplii.



A group of male *M. jakora*.

Photo: Johannes Graf

I gave some of these offspring to fellow breeder Erwin Binner at the IRG Annual General Meeting 2018. Through various experiments we both found out that it was not the humic substances themselves, but the lowered pH value due to the acids contained in them that was the reason for the successful development of the eggs. pH 4 to at most pH 5 was the value at which the eggs developed successfully. The fish breeding colleagues that didn't believe us and to whom we gave young fish, all failed to successfully reproduce this fish. Therefore, *Melanotaenia jakora*, as it is now called, has so far, not spread far in the aquarium hobby.

So, Erwin and I bred this beautiful species over the years until the fourth generation of offspring when suddenly an interesting thing occurred: the larvae hatched in normal, but relatively soft water with up to pH 6.5. Apparently, domestication has already taken place here. The current offspring generation hatched at pH 6.5 without major losses.

Melanotaenia jakora should preferably be kept in a species tank, as the species is not very assertive towards other rainbowfishes. In an aquarium lit with moderate lighting from the front and setup with wood and some plants, these fish display their iridescent colours particularly beautifully. The water should be soft and rich in humic substances and the temperature around 25 °C. The courtship displays of the males are wonderful to watch. As with all smaller rainbowfish species, the life expectancy of the individual is about two years. Therefore, you should either regularly add offspring or ensure that young animals can grow up in the tank by furnishing the tank with a protected area (floating plants, java moss up to the surface).

In conclusion, I hope that this beautiful species can be established in the hobby in the long term despite its somewhat special requirements. The destruction of habitats in New Guinea continues and it is becoming increasingly important to conserve rainbowfish species in the hobby for the future.



Newly hatched fry of *M. jakora*.

Photo: Johannes Graf

